

Animal & Plant Cells

Answer sheet · 22 marks total

Section A — Label the animal cell

1. Nucleus 2. Mitochondria 3. Ribosomes 4. Cell membrane 5. Cytoplasm

[1 mark each, 5 marks total]

Section B — Label the plant cell

1. Cell wall 2. Cell membrane 3. Nucleus 4. Chloroplast 5. Permanent vacuole 6. Cytoplasm

[1 mark each, 6 marks total]

Section C — Function matching

C1. Nucleus — contains the genetic material (DNA), controls the cell's activities (1).

C2. Mitochondria — site of respiration, releases energy (1).

C3. Ribosomes — where proteins are made (1).

C4. Chloroplast — site of photosynthesis (1).

C5. Cell wall — strengthens/supports the cell (1).

Section D — Animal vs plant

D1. Any two of: cell wall, chloroplasts, permanent vacuole (1 mark each, 2 marks total).

D2. Plant cells need a rigid shape / structural support because plants cannot move around (1); the cell wall gives the cell this rigidity/strength, which an animal cell does not need (1).

Section E — Bacterial cells

E1. Plasmid(s) (1).

E2. In a bacterial cell the genetic material is a single DNA loop free in the cytoplasm (not in a nucleus); in an animal cell it is enclosed in a nucleus (1).

Marking note: accept any scientifically correct equivalent wording throughout. For labelling sections, accept minor spelling errors as long as the structure is clearly identifiable.